SE LAB6

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(venv) PS C:\Users\shant\Desktop\sem5\SE_Lab_PES2UG24CS456\lab6> python -m coverage run --source=parking_fee -m pytest CodeCoverageTest_parking_fee.py
============================= test session starts ==============================
platform win32 -- Python 3.13.7, pytest-9.2.0, pluggy-1.7.0
rootdir: C:\Users\shant\Desktop\sem5\SE_Lab_PES2UG24CS456\lab6
collected 17 items

CodeCoverageTest_parking_fee.py .................

============================== 17 passed in 0.29s ==============================
(venv) PS C:\Users\shant\Desktop\sem5\SE_Lab_PES2UG24CS456\lab6> python -m coverage report -m
Name             Stmts   Miss  Cover   Missing
----------------------------------------------
parking_fee.py      34      0   100%
----------------------------------------------
TOTAL               34      0   100%
(venv) PS C:\Users\shant\Desktop\sem5\SE_Lab_PES2UG24CS456\lab6>
```

```
(venv) PS C:\Users\shant\Desktop\sem5\SE_Lab_PES2UG24CS456\lab6> python -m coverage run --branch --source=parking_fee -m pytest BranchCoverageTest_parking_fee.py
============================= test session starts ==============================
platform win32 -- Python 3.13.7, pytest-9.2.0, pluggy-1.7.0
rootdir: C:\Users\shant\Desktop\sem5\SE_Lab_PES2UG24CS456\lab6
collected 17 items

BranchCoverageTest_parking_fee.py .................

============================== 17 passed in 0.57s ==============================
(venv) PS C:\Users\shant\Desktop\sem5\SE_Lab_PES2UG24CS456\lab6> python -m coverage report -m
Name             Stmts   Miss Branch BrPart  Cover   Missing
------------------------------------------------------------
parking_fee.py      34      0     24      0   100%
------------------------------------------------------------
TOTAL               34      0     24      0   100%
(venv) PS C:\Users\shant\Desktop\sem5\SE_Lab_PES2UG24CS456\lab6>
```

Parking Fee Application — Test Coverage Report

This report documents the test coverage results for the parking-fee application. The assessment follows the Lab 6 evaluation rubric: code coverage analysis and report (2 marks), branch coverage analysis and report (2 marks), and commits and test cases achieving greater than 90% coverage (6 marks).

1. Objective

The objective was to evaluate how much of **parking_fee.py** is executed by the automated tests and whether the tests exercise the alternative outcomes of conditional decisions. The Python **pytest** framework was used to execute the tests, and the **coverage** package was used to collect line and branch coverage.

2. Test Environment and Procedure

• Environment: Windows with Python 3.13.7, pytest 9.2.0, and coverage installed in a virtual environment.

• Application under test: parking_fee.py.

• Code coverage test file: CodeCoverageTest_parking_fee.py.

• Branch coverage test file: BranchCoverageTest_parking_fee.py.

• Line coverage was measured by running coverage with pytest and then viewing the coverage report.

• Branch coverage was measured by enabling the --branch option and generating a separate report.

3. Code Coverage Analysis

The code-coverage test suite completed successfully with **17 tests passed**. The coverage report showed that all 34 executable statements in parking_fee.py were executed, with zero missed statements. The resulting statement coverage was **100%**, exceeding the rubric requirement of greater than 90%.

Metric	Result	Interpretation
Tests executed	17 passed	All tests in the suite passed
Statements	34	Executable statements measured
Missed statements	0	No statements left unexecuted
Code coverage	**100%**	Meets and exceeds the 90% target

4. Branch Coverage Analysis

The branch-coverage test suite also completed successfully with **17 tests passed**. The report identified 24 branches and zero missing branches. There were no partially covered branches, and the resulting branch coverage was **100%**. This indicates that the tests exercised both outcomes of the decision branches measured by the coverage tool.

Metric	Result	Interpretation
Tests executed	17 passed	All tests in the suite passed
Statements	34	All statements were executed
Branches	24	Conditional outcomes measured
Missing / partial branches	0 / 0	No uncovered or partially covered branches
Branch coverage	**100%**	Meets and exceeds the 90% target

5. Test Case Coverage

The test cases exercise the main behaviours and decision outcomes in the parking-fee module. The test suite covers normal values, boundary values, and error conditions across the three application functions.

Function	Behaviours exercised
parking_fee()	Vehicle rates, short/free stays, duration rounding, invalid vehicle, negative duration, EV discount eligibility, truck EV exception, lost-ticket surcharge, and combined EV/lost-ticket behaviour.
find_slot()	Full lot, disabled parking priority, truck basement allocation, and ordinary floor allocation.
monthly_pass()	Bike and car prices, resident discounts, non-resident pricing, and unsupported vehicle errors.

6. Git Commit Evidence

The two coverage-related commits were created and pushed to the repository's main branch. A subsequent commit added the application and starter test files.

Commit	Message	Purpose
8c9246c	Add parking fee code coverage tests for lab-6	Adds code-coverage test file and screenshot
25575ca	Add parking fee branch coverage tests for lab-6	Adds branch-coverage test file and screenshot
fc9d345	Add parking fee application and starter test files	Adds application, starter tests, and README

7. Conclusion

The parking-fee application achieved 100% statement coverage and 100% branch coverage in the recorded runs, with all 17 tests passing in each suite. The results exceed the lab's 90% coverage target. The submitted test files and screenshots provide evidence for both coverage analyses, and the Git history records the coverage test additions. High coverage improves confidence that the tested statements and decision outcomes are exercised, although it does not by itself prove the absence of all defects.